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Experiment Name : **a) ANALYZE THE TREND OF DATA SCIENCE JOB POSTINGS OVER THE LAST DECADE**

b) ANALYZE AND VISUALIZE THE DISTRIBUTION OF VARIOUS DATA SCIENCE ROLES (DATA ANALYST, DATA ENGINEER, DATA SCIENTIST, ETC.) FROM A DATASET

c) CONDUCT AN EXPERIMENT TO DIFFERENTIATE STRUCTURED, UNSTRUCTURED AND SEMI-STRUCTURED DATA BASED ON DATASETS GIVEN

d) CONDUCT AN EXPERIMENT TO ENCRYPT AND DECRYPT GIVEN SENSITIVE DATA

In [3]:
```
import pandas as pd

import matplotlib.pyplot as plt

data = {'Year': list(range(2010, 2021)),

'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400,
4200]}

df = pd.DataFrame(data)

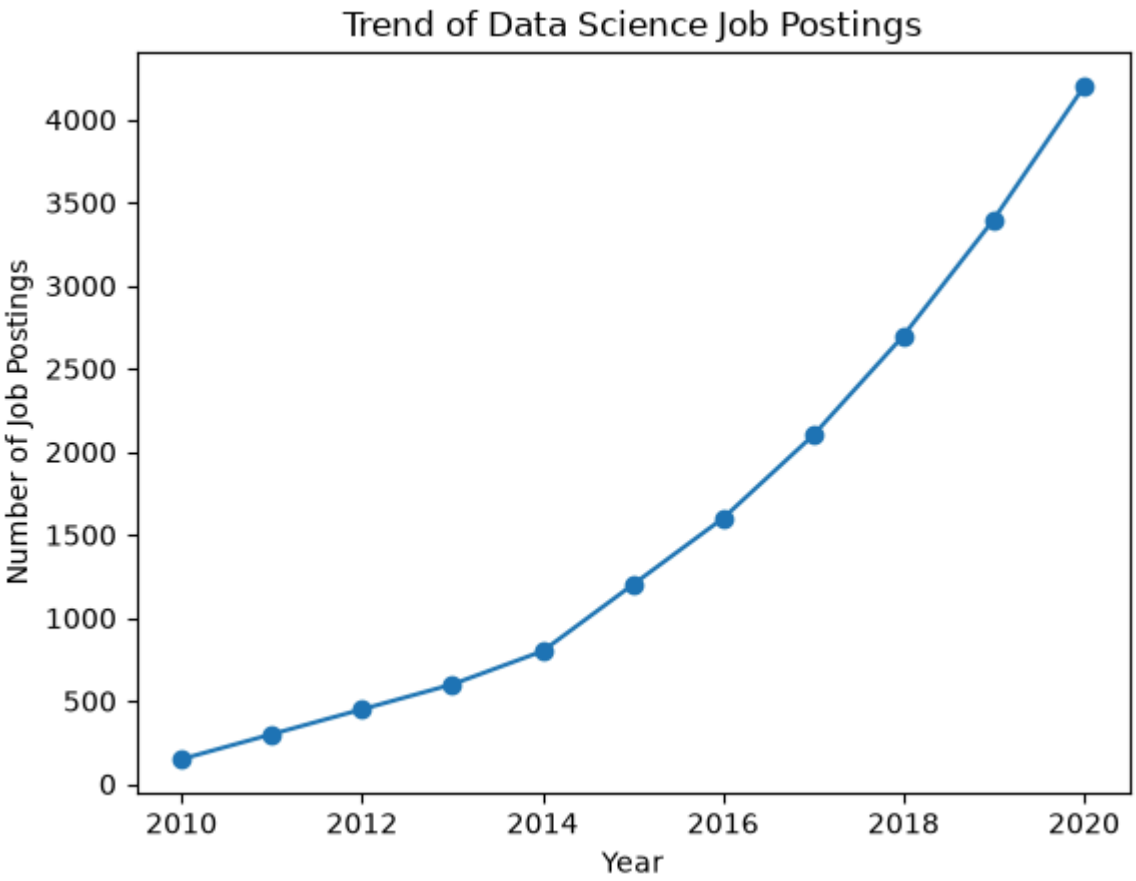
plt.plot(df['Year'], df['Job Postings'], marker='o')

plt.title('Trend of Data Science Job Postings')

plt.xlabel('Year')

plt.ylabel('Number of Job Postings')

plt.show()
```



In [4]:
```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer',
'Business Analyst']

counts = [300, 500, 450, 200, 150]

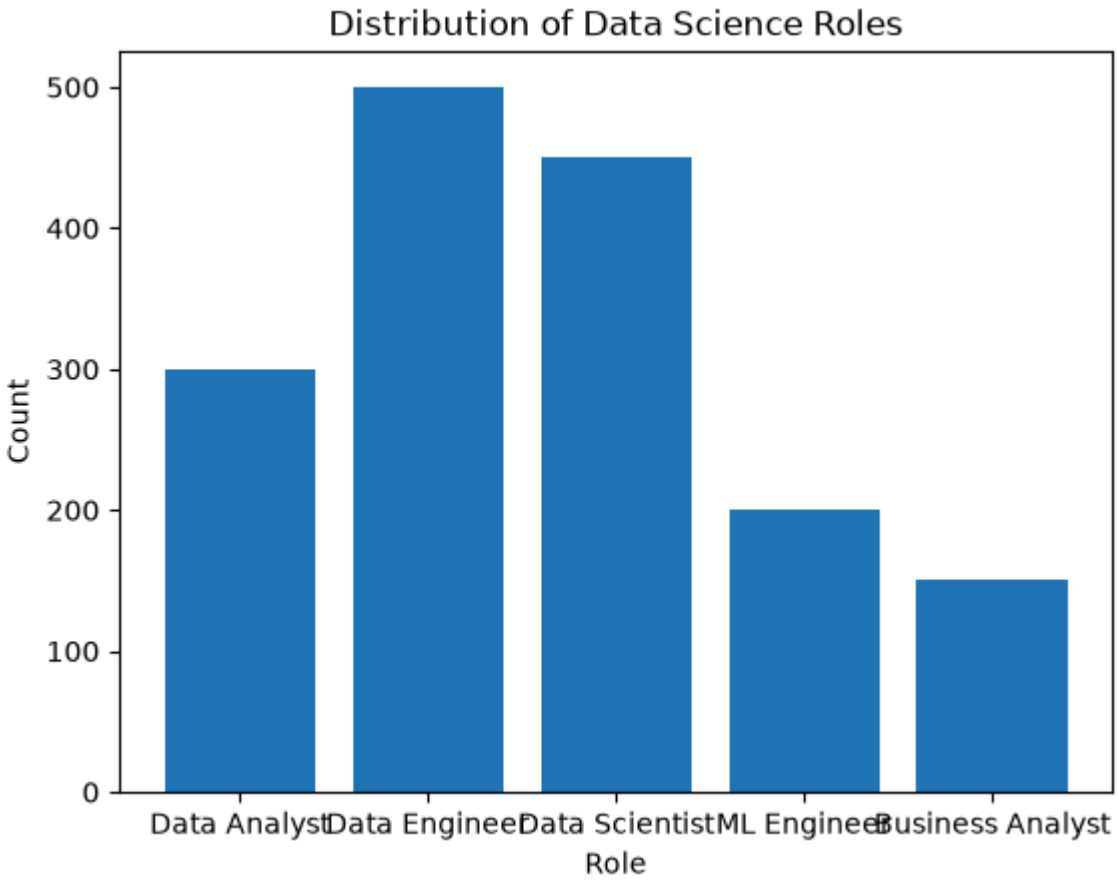
plt.bar(roles, counts)

plt.title('Distribution of Data Science Roles')

plt.xlabel('Role')

plt.ylabel('Count')

plt.show()
```



In [5]:
```
# Structured data example

structured_data = pd.DataFrame({

'ID': [1, 2, 3],

'Name': ['Alice', 'Bob', 'Charlie'],

'Age': [25, 30, 35]

})

print("Structured Data:\n", structured_data)

# Unstructured data example

unstructured_data = "This is an example of unstructured data. It can be a
piece of text, an image, or a video file."

print("\nUnstructured Data:\n", unstructured_data)

# Semi-structured data example (JSON)

semi_structured_data = {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height':
165, 'Weight': 68}}

print("\nSemi-structured Data:\n", semi_structured_data)
```

```
Structured Data:
    ID     Name  Age
0   1    Alice   25
1   2      Bob   30
2   3  Charlie   35

Unstructured Data:
 This is an example of unstructured data. It can be a piece of text, an
image, or a video file.

Semi-structured Data:
 {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
```

In [7]:

```
# Generate key and encrypt data

from cryptography.fernet import Fernet

key = Fernet.generate_key()

f = Fernet(key)

token = f.encrypt(b"Rajalakshmi Engineering College")

token

b'...'

f.decrypt(token)

b'Rajalakshmi Engineering College'

key = Fernet.generate_key()

cipher_suite = Fernet(key)

plain_text = b"Rajalakshmi Engineering College."

cipher_text = cipher_suite.encrypt(plain_text)

# Decrypt data

decrypted_text=cipher_suite.decrypt(cipher_text)

print("Original Data:", plain_text)

print("Encrypted Data:", cipher_text)

print("Decrypted Data:", decrypted_text)
```

```
Original Data: b'Rajalakshmi Engineering College.'
Encrypted Data:
b'gAAAAABqjtbI8MIFy1BvapPTld8vqmhbPCcn1Oav6TiAI1lmUzJ5xTaZdl82o1uOQNd2p3yBFA
-KWgkxA7f-h7t2U8-Z24E2xZhDybAISWGUelbheMoTw0tVRElkjCoaUvzB9d5seh8P'
Decrypted Data: b'Rajalakshmi Engineering College.'
```